

Aid Preferences Among Subsistence Farmers in the DRC*

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Abstract

This paper examines beneficiary preferences over the design of a bundled resilience program and how gender-related constraints shape them. Using a conjoint experiment that simulates aid cuts, it estimates willingness to pay for key assistance attributes and compares preferences among matched male and female respondents within the same household. The design allows for the analysis of heterogeneity and intra-household differences in choices. The findings show that beneficiaries primarily view assistance as a means of meeting immediate needs rather than as a financial instrument for productive investment. Both men and women strongly prefer cash transfers, particularly cash in hand, while complementary components such as training and seeds are valued less. Men place greater value on immediate transfers, whereas women place greater value on assistance during the lean season, consistent with consumption smoothing concerns. Preference heterogeneity is more pronounced along gender-related dimensions than along standard vulnerability indicators. Intra-household disagreement is substantial, with partners choosing different packages in about one-third of scenarios, though disagreement is lower in households with stronger social cohesion, higher trust, and greater male engagement in care work.

Keywords: Gender, Preferences, Conditional assistance, Conjoint experiment

JEL Codes: O12, D91, J16

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1 Introduction

Conditional transfers, bundled and cash-plus interventions, and graduation programs typically combine cash or in-kind transfers with complementary components such as labor requirements, training, productive inputs, and other forms of livelihood support (Pruce et al., 2025; Sedlmayr et al., 2020; Leight et al., 2024). These design choices interact with the binding economic and social constraints beneficiaries face—such as liquidity and credit constraints or gender norms—so households are likely to differ systematically in their preferences over program features, in their take-up, and in the welfare gains they derive from assistance (Vera-Cossio, 2022; Kansikas et al., 2025; Cortés et al., 2026; Gideon, 2019; Fisher et al., 2017; Hasan, Amer, 2010; Anagol et al., 2025; Finkelstein and Notowidigdo, 2019; Ko and Moffitt, 2024).¹ Yet most decisions about these design alternatives are made with limited input from potential beneficiaries (Rabinovich and Diepeveen, 2015; Yeboah et al., 2019; Doocy and Tappis, 2017).

Evidence on how beneficiaries value alternative program designs—and how economic and social constraints, including gender, shape these preferences—remains limited. Most existing studies isolate a single attribute, most prominently transfer modality, abstracting from the multidimensional nature of assistance architecture (Hirvonen and Hoddinott, 2021; Satapathy et al., 2022; Gentilini, 2016). Moreover, preferences are also typically elicited from a single household respondent or in samples underpowered to identify gender-specific valuations (Alzua et al., 2017; Dillon and Mensah, 2024; Harris and van der Windt, 2023). As a result, little is known about whether male and female members of the same household value features such as complementary components, recipient designation, or delivery mechanisms differently, or whether such divergence generates intra-household tension in uptake and satisfaction (Guilbert and Pierotti, 2016; Ambler et al., 2021; Bakhtiar et al., 2025; Ambler et al., 2025).

¹This is consistent with a growing literature documenting heterogeneous treatment effects of cash transfers and related programs. For reviews and meta-analyses, see Kabeer and Waddington (2015), Bastagli et al. (2019), and Crosta et al. (2024).

We implement a conjoint experiment among matched male and female respondents within the same households participating in the Joint Resilience Program in South Kivu, Democratic Republic of the Congo, to contribute to this gap. Respondents completed identical choice tasks in private interviews, allowing a direct comparison of gender specific trade offs over transfer size, modality, timing, complementary components, recipient gender, and targeting approach. The experimental design embeds hypothetical reductions in transfer value, enabling the estimation of willingness to pay measures that simulate budget cuts. It also allows the assessment of intra household differences in both choices and underlying attribute valuations, as well as their potential determinants.

The setting provides a unique opportunity to examine how economic and social constraints shape program preferences in a polycrisis context, where these constraints may be especially pronounced for women. South Kivu is characterized by recurrent conflict and climate-related shocks that intensify liquidity pressures, increase labor and care burdens, and heighten insecurity (Trako et al., 2024; WFP, 2025). These conditions reinforce pre-existing inequalities in access to labor and financial markets, potentially widening gender differences in the valuation of income, flexibility, and program requirements. Combined with rich household survey data, the context allows us to examine how gendered preferences vary with socioeconomic status, exposure to shocks, time allocation, and measures of agency and norms.

The within-household design enables a direct assessment of intra-household preference divergence. Because male and female respondents face identical choice tasks, we can observe whether spouses select different assistance packages and quantify the extent of disagreement across decisions. Beyond comparing attribute coefficients, we classify alternatives into typologies of assistance packages—such as liquidity-oriented versus investment-oriented bundles—and examine whether respondents (within the same household) systematically favor different types of support. This design further allows us to identify the socioeconomic and vulnerability factors associated with intra-household disagreement in program choices.

We find that beneficiaries primarily view assistance as a mechanism to address immediate needs rather than as a tool for productive investment. Both male and female respondents show strong preferences for cash transfers relative to in-kind assistance, particularly for cash in hand: respondents are willing to forgo roughly USD 87 (men) and USD 67 (women) of the transfer to receive cash in hand, compared with about USD 41 and USD 24, respectively, for mobile money (all $p < 0.01$). Complementary components such as training and seeds are also valued, though to a lesser extent, with men forgoing about USD 76 and women about USD 52 of transfer value for training (both $p < 0.01$). Timing preferences reinforce this pattern: men place greater value on immediate assistance, while women place relatively more value on transfers delivered during the lean season, consistent with stronger consumption smoothing concerns. Preference patterns are broadly similar across gender but differ in magnitude. Women display stronger preferences for direct cash modalities and for being designated as the transfer recipient, forgoing about 50% of the transfer to direct assistance to women, whereas men are statistically indifferent to recipient gender. The embedded aid-cut variation reveals a similar asymmetry: women’s utility falls sharply as cuts deepen—a 40% reduction lowers utility by $\beta = -0.27$ ($p < 0.05$) and a 60% reduction by $\beta = -0.21$ ($p < 0.01$)—whereas the corresponding effects for men are statistically indistinguishable from zero, consistent with women bearing a larger share of the household’s consumption-smoothing burden. Heterogeneity is more pronounced along gender-related dimensions, with differences in female agency, time burdens, and intra-household labor allocation associated with variation in the valuation of program attributes, while vulnerability and shock exposure play a more limited role. Land tenure shapes men’s preferences but not women’s: among men, preferences for complementary components and cash modalities are evident only in land-owning households, whereas women exhibit the same preference structure regardless of land ownership, suggesting that men’s preferences are more sensitive to productive assets than women’s.

We also document meaningful intra-household differences in preferences. Partners select

different assistance packages in roughly one-third of choice scenarios—about two decisions per household across the experiment—and nearly three quarters (74%) of households exhibit at least one disagreement. Women are also about 3 percentage points more likely than men to choose packages that direct the transfer to their own gender. Disagreement is lower in households with stronger social cohesion, higher levels of trust, and greater male engagement in work and care activities, suggesting that household dynamics and social norms shape how assistance is valued within the household.

We make several contributions. First, we contribute methodologically by extending the application of discrete choice experiments to the study of beneficiary preferences in social assistance. While DCEs are widely used in health, environmental, and consumer economics (Regassa et al., 2023; Dunsch and Velényi, 2026), their application to program design among intended beneficiaries remains limited. The closest precedents are Rao et al. (2020), who use a mixed-logit DCE to study mothers’ preferences over five attributes of a maternal-health conditional cash transfer in India, and Yeboah et al. (2019), who apply a similar approach to conditional cash transfers in Ghana with attention to conditionality, targeting, and payment method. Abreh et al. (2024) examine preferences over transfer modality in Ghana’s school feeding program. Although these studies provide valuable insights, they each examine a limited set of attributes and rely on aggregate, single-respondent designs. We extend this line of work in three ways: (i) a matched within-household design that identifies gender-specific valuations rather than population averages; (ii) an embedded transfer-cut variation that simulates an aid cut directly in the choice task rather than holding cash levels fixed; and (iii) a multidimensional design that varies six program features simultaneously, allowing the joint estimation of trade-offs across modality, timing, scope, recipient, and targeting—dimensions previously studied only in isolation.

Second, we contribute to the broader literature on beneficiary preferences and program design. Despite the prominence of beneficiary choice in policy discussions, systematic evidence on how recipients evaluate alternative program configurations remains scarce. Existing

literature tends to examine individual design elements of social assistance programs in isolation. The most prominent strand focuses on transfer modality, comparing cash, in-kind transfers, or digital payments (see, for instance, [Khera, 2014](#); [Hidrobo et al., 2014](#); [Hirvonen and Hoddinott, 2021](#); [Shapiro, 2019](#); [Ambler et al., 2025](#); [Das and Sethi, 2023](#)). Other studies analyze recipient identity within the household ([Haushofer and Shapiro, 2016](#)), the timing and tranching of transfers ([Kansikas et al., 2025](#); [Haushofer and Shapiro, 2016](#)), or transfer size ([Haushofer and Shapiro, 2016](#)). While these studies provide valuable insights into specific dimensions of program design, they typically evaluate each attribute separately. By analyzing these dimensions jointly within a unified experimental framework, we estimate the relative importance of distinct design features and provide evidence on how beneficiaries evaluate complete program configurations rather than single attributes.

A relatively understudied dimension of our design concerns whether beneficiaries prefer simple cash transfers or bundled programs that combine transfers with complementary components such as training, productive inputs, or livelihood support.² A small but growing literature incorporates beneficiary perspectives. [Allen et al. \(2024\)](#) find that when households choose between cash transfers and economic inclusion programs, many prefer the latter, though preferences are heterogeneous. [Rao et al. \(2020\)](#) shows that mothers value conditional cash transfers partly because the conditions facilitate access to child health services, and [Skovdal et al. \(2014\)](#) document that communities accept conditionality more readily when requirements are perceived as fair and aligned with locally valued activities.

Third, we contribute to the literature on heterogeneity in beneficiary preferences by providing direct evidence on gender specific differences in program design valuations and by examining how these differences vary across a rich set of characteristics and constraints. Existing evidence shows that preferences for delivery modalities vary with economic conditions and constraints: while some households favor cash for the flexibility it provides ([Hidrobo](#)

²Such designs are common in public works schemes, conditional cash transfer programs, and graduation-style interventions targeting extreme poverty ([Hirvonen et al., 2023](#); [Banerjee et al., 2022](#); [Bossuroy et al., 2022](#); [Bandiera et al., 2017](#); [Banerjee et al., 2015](#)).

et al., 2014), others exposed to price volatility (Hirvonen and Hoddinott, 2021) or facing self-control concerns (Khera, 2014) may prefer in-kind transfers to stabilize consumption.

We further contribute to the literature by providing direct evidence on intra-household divergence in program design preferences. Our approach differs from Abushama et al. (2025), who study binary modality preferences using survey-reported choices. By contrast, our conjoint design varies multiple program features simultaneously and allows us to identify not only whether male and female respondents differ, but also the intensity of those differences and systematic patterns of divergence across complete assistance packages. This approach enables us to recover preference trade-offs across program attributes and to examine how intra-household differences emerge when beneficiaries evaluate realistic policy bundles rather than isolated design elements.

The remainder of the paper proceeds as follows. Section II describes the study context and the Joint Resilience Program. Section III presents the experimental design, sample, and data. Section IV outlines the analytical framework and estimation strategy. Section V reports the main results on preferences, willingness to pay, heterogeneity, and intra-household differences. Section VI concludes with policy implications.

2 Study Setting

2.1 The Democratic Republic of Congo

The Democratic Republic of the Congo (DRC) remains among the world’s poorest economies, with an estimated 85.3% of the population—roughly 82 million people—living below the international poverty line of USD 3.00 per day in 2020 (World Bank, 2025a). Estimates suggest that close to 50 million Congolese were living in extreme poverty in 2024, the second largest such population in Sub-Saharan Africa (World Bank, 2025a). Persistent fragility is reflected in the country’s ranking of 171st out of 193 countries on the 2025 Human Development Index, placing it firmly within the low human development group (UNDP, 2025).

DRC represents a stark case of polycrisis, in which multiple, mutually reinforcing shocks interact to intensify vulnerability. Recurrent extreme weather events pose a significant threat to national food systems, particularly given that nearly 70% of the population relies on rainfed agriculture for its livelihoods ([USAID, 2023](#)). Combined with conflict, displacement, and broader economic disruptions, these pressures have driven the country into a protracted food crisis, leaving it with the largest absolute number of food-insecure people globally ([GRFC, 2023](#)). An estimated 24.8 million people face high levels of acute food insecurity ([IPC, 2025a](#)), while approximately 5.28 million individuals are internally displaced (IDPs), the second largest internally displaced population in Africa after Sudan ([IDMC, 2024](#)).

Conflict, displacement, and weather shocks have disproportionately affected women and girls. Female-headed households are 7% more likely to experience moderate or severe food insecurity than male-headed households (79% versus 74%) ([WFP, 2025](#)), and 1.7 million out of the 4.2 million malnourished people are pregnant or lactating women ([IPC, 2025b](#)). This disproportionate burden borne by women reflects entrenched gender imbalances in economic and social participation, limited voice in public decision making, and persistent sociocultural inequalities ([Trako et al., 2024](#)). DRC ranks 151st out of 179 countries on the 2021 Gender Inequality Index, which measures disparities in reproductive health, empowerment, and labor market participation ([UNDP, 2022](#)). These disparities also reflect on earnings and land ownership despite women’s relatively high labor force participation (63.1% in 2022), particularly in agriculture (64.8% of working women; [ILO, 2021](#)). Gender based violence is also pervasive, with 47% of women aged 15 to 49 having experienced intimate partner violence at least once ([VAW-IAWGED, 2021](#)).

A polycrisis context, such as that of the DRC, marked by persistent inequality and the disproportionate impact of shocks on women, offers a particularly relevant setting in which to examine how multiple, interacting stressors shape preferences for assistance and its design.

2.2 The Joint Resilience Program

The World Food Programme (WFP), the Food and Agriculture Organization (FAO), and the United Nations Children’s Fund (UNICEF), with financial support from the German Federal Ministry for Economic Cooperation and Development (BMZ), jointly implemented the Joint Resilience Program titled Enhanced Community Structures and Organizations to Promote Gender Equity, Peace, and Social Cohesion in North and South Kivu between 2020 and 2023. The program aimed to strengthen the socioeconomic resilience of smallholder farmers and vulnerable populations through a multi-component approach combining agricultural and livelihood interventions, community-based nutrition services, water, sanitation, and hygiene (WASH) services, and social cohesion and peacebuilding activities.

The program targeted approximately 180,000 beneficiaries. The livelihood component included the provision of agricultural inputs to improve production and productivity, targeted livelihood training, the establishment of Village Savings and Loan Associations, and a key component of cash for work (CFW) designed to alleviate household economic constraints. Agricultural inputs included biofortified or improved seeds intended to strengthen crop yields and enhance resilience to climatic variability. Training modules focused on improving agricultural practices and reducing post-harvest losses, covering quality control of crops, pre-harvest methods, sorting, drying, and basic processing techniques. The cash for work component included rehabilitation of feeder roads to improve connectivity between remote rural areas and more densely populated cities, as well as restoration of degraded land to increase agricultural productivity and preserve natural resources ([WFP et al., 2024](#)).

The targeting of input provision and training, as well as other program components, was conducted through farmer organizations supported by the resilience project and followed established eligibility criteria. To participate, individuals were required to be core members of a farmer organization supported by the project, maintain continuous residence in the targeted area for the duration of the activity, and, in the case of the cash for work component, live within approximately five kilometers of the rehabilitation site for feeder road works in

order to ensure regular participation. The targeting also reflected the program priorities related to vulnerability and gender inclusion, with a recommended minimum participation rate of 60% of women. The duration of the engagement was not tied to a fixed agricultural cycle, but depended on the scope and estimated cost of the rehabilitation activity and the number of beneficiaries mobilized for implementation.

Participating households in the cash for work component received a daily transfer of approximately USD 3.5 for 20 to 22 working days per month, corresponding to a monthly payment of USD 70 to 77 in nominal terms. Transfers were typically disbursed on a monthly basis. This amount represents approximately 94% of the 2023 statutory monthly minimum wage of CDF 183,950,³ or approximately USD PPP 194 per month in purchasing power parity terms,⁴ and covers approximately 58% of the per capita poverty threshold for an average household of 5 members.⁵

3 Study Design

Assistance packages bundle multiple design choices that jointly shape budgets, effectiveness, and beneficiary welfare. Given that these features are implemented and experienced together, the trade-offs they generate must be evaluated simultaneously rather than one at a time. A discrete choice experiment is well-suited to this setting: it identifies marginal rates of substitution across package attributes, providing a coherent framework for analyzing the multidimensional design of assistance interventions. The DCE further allows us to elicit will-

³Equivalent to USD 78 at the 2023 average market exchange rate of 2,360 CDF/USD. We computed it as USD 73.5 / USD 78 $\approx$ 94%, where USD 73.5 is the midpoint of the monthly transfer range and USD 78 is CDF 183,950 converted at the 2023 average market exchange rate of 2,360 CDF/USD ([ILO, 2025](#)).

⁴Applying the World Bank’s 2023 PPP conversion factor of 950 CDF per international dollar, we computed it as CDF 183,950 / 950 $\approx$ USD PPP 194 for the equivalent of the minimum wage; the monthly transfer of USD 70–77 nominal corresponds to approximately USD PPP 174–191 (midpoint USD PPP 183), obtained by multiplying the nominal USD value by the ratio of the market exchange rate to the PPP conversion factor (2,360 / 950 $\approx$ 2.48) ([World Bank, 2025c](#)).

⁵Relative to the international poverty line of USD 3.00 per person per day in 2021 PPP ([World Bank, 2025b](#)). We computed it as (USD PPP 183 / 30 days) / 5 members $\approx$ USD PPP 1.22 per person per day, or 41% of the USD 3.00 per capita daily line if averaged over the month; or, equivalently, USD PPP 8.69 per household per day (USD 3.5 $\times$ 2.48) divided across 5 members yields USD PPP 1.74 per person per day on working days, or roughly 58% of the line.

ingness to pay for specific attributes under scenarios in which the overall value of assistance is reduced, reflecting the fiscal constraints currently facing development programs.

3.1 Sample and Data

We conduct a discrete choice experiment in South Kivu, DRC, among subsistence farmers exposed to conflict and climate variability who are members of farmer organizations participating in the Joint Resilience Program. The experiment was implemented between August and September 2023 as part of a joint assessment by WFP and the World Bank’s Development Impact Monitoring and Evaluation unit, examining gender, social cohesion, and resilience outcomes.

Data were collected through computer assisted personal interviews averaging approximately two hours. Enumerators underwent a two week training, including piloting in comparable communities. Fieldwork was monitored through daily quality checks and back checks on a random 20% subsample of households. The study received ethical approval from Solutions IRB on July 4, 2023, and all participants provided informed consent.

The final sample comprises 1,611 households across 106 villages in Mwenga and Walungu. Households were selected proportionally to village population size using a random walk procedure. Within each household, the survey was administered separately to the female and male household head or, when unavailable, to a primary male decision maker. The analytical sample, therefore, includes 3,222 respondents, enabling the estimation of gender disaggregated preferences and intra-household heterogeneity in choices. Appendix Table [A1](#) contains details of the sample by village.

Table 1 presents descriptive statistics for households and respondents. Households are relatively large, with an average size of 7 members, and the majority (87%) consist of a married or cohabiting couple. 6% of households are classified as internally displaced persons (IDPs), while 28% are returnees. Most households (69%) report owning or renting land.

Food insecurity⁶ affects 39% of households, and exposure to shocks is widespread. 65% report having adopted crisis or emergency coping strategies, 23% were affected by conflict in the past 12 months, and 81% experienced a climate shock over the same period.

At the individual level, the average age of female respondents is 39 years, while male respondents are slightly older with an average age of 44 years. Primary education attainment remains limited, with 37% of women and 34% of men having attended or completed primary school. Employment in the past 12 months is reported by 25% of women and 28% of men, whereas involvement in agricultural activities is common among both groups (76% of women and 74% of men). Among those reporting earnings, men earn on average CDF 131,739 per month compared to CDF 80,151 for women, indicating a substantial gender gap in monetary income.

3.2 Conjoint experiment

Discrete choice experiments are grounded in random utility theory, which models goods and services as bundles of attributes from which individuals derive utility (Lancaster, 1966). Under this framework, respondents choose the alternative that maximizes their utility. We extend this approach to social assistance by implementing a discrete choice experiment that elicits beneficiaries’ preferences over alternative assistance packages. Identification relies on randomized variation in attribute levels across choice tasks, enabling estimation of the marginal utility of each feature (Hensher and Greene, 2003; Train, 2009).

Respondents participating in the Joint Resilience Program (JRP) were repeatedly asked to choose between two feasible assistance packages that varied across predefined design dimensions. Because the objective is to elicit preferences over program features rather than participation decisions, no opt-out option was provided.

Attributes were selected based on three criteria: relevance in the literature on transfer design and conditional or graduation programs, operational feasibility confirmed through

⁶Food insecurity is measured using the Food Consumption Score (FCS) and defined as an FCS below 28, in line with WFP classification thresholds (WFP, 2008).

Table 1: Descriptive Statistics of Respondent and Household

	Male			Female		
	Mean	SD	N	Mean	SD	N
Age	44	15.33	1,611	38.91	14.27	1,611
Attended/completed primary	0.34	0.47	1,611	0.37	0.48	1,611
Employed in the last 12m	0.28	0.45	1,611	0.25	0.43	1,611
Involved in agriculture	0.74	0.44	1,611	0.76	0.43	1,611
Monthly income/salary (CDF)	131,739	328,135	397	80,151	335,515	373
Household						
	Mean		SD		N	
Household size	7.14		2.65		1,611	
Married or cohabiting	0.85		0.35		1,611	
Owns or rents land	0.69		0.46		1,611	
Food insecure (FCS < 28)	0.39		0.49		1,611	
IDP household	0.06		0.24		1,611	
Returnee household	0.28		0.45		1,611	
Crisis/emergency coping	0.64		0.48		1,611	
Conflict shock (last 12m)	0.10		0.30		1,611	
Climate shock (last 12m)	0.30		0.46		1,611	

Notes: Means and standard deviations computed on the analytic sample of 1,611 households interviewed in South Kivu (Mwenga and Walungu) between August and September 2023. The upper panel reports respondent-level statistics for matched male and female respondents within each household; the lower panel reports household-level statistics. Income statistics in the upper panel use the subsample of respondents reporting positive earnings (N = 397 male, 373 female); all other variables use the full sample. Food insecurity follows the Food Consumption Score (FCS) classification with $FCS < 28$; coping behavior from the Livelihood Coping Strategy Index (LCSI). *Source:* Authors' calculations on the WFP Gender, Social Cohesion, and Resilience survey, South Kivu, 2023.

consultation with WFP’s DRC country office, and piloting to ensure clarity and cognitive manageability. The final design includes six attributes: transfer value, timing, program scope, delivery modality, recipient within the household, and targeting method. **Table 2** presents the attributes and levels used in the experiment.

Table 2: Assistance attributes and attribute levels used in the choice experiment

Attribute	Definition	Attribute levels
Transfer value	Represents the value expected to be received monthly for participation in the FFA activities.	(1) 160,000 CDF, (2) 130,000 CDF, (3) 100,000 CDF, (4) 70,000 CDF
Timing	Whether beneficiaries have preferences to delay transfer to match particular timeframes and needs.	(1) Two weeks from now, (2) Starting the lean season, (3) Starting the planting season
Scope	Whether beneficiaries would prefer or sacrifice transfer value for more comprehensive approaches rather than just transfers.	(1) Only transfer, (2) Transfer + training, (3) Transfer + seeds
Modality	The modality in which beneficiaries prefer to receive assistance.	(1) In kind, (2) Cash in hand, (3) Cash via mobile money
Recipient	The person in the household that the beneficiary would rather choose as the recipient of the assistance.	(1) Female, (2) Male
Targeting method	The method the beneficiaries would prefer to conduct the selection of beneficiaries.	(1) Community driven, (2) Data driven

Notes: Each respondent completed six choice tasks, each presenting two assistance packages constructed by combining one level from each of the six attributes. Transfer values reflect the average monthly cash-for-work payment under the standard programming (160,000 CDF) and reductions of 20%, 40%, and 60%. The full design has 192 possible profiles; a D-optimal fractional factorial design generated 48 choice pairs split into eight blocks of six tasks each. *Source:* Authors’ construction based on the existing literature and consultation with WFP’s DRC country office.

The *transfer value* attribute represents the monthly payment received for participation in cash for work activities. The highest level, 160,000 Congolese Francs, corresponds to the average monthly transfer under standard programming. The remaining levels reflect reductions of 20%, 40%, and 60%, allowing us to simulate trade offs between transfer size and other program features in a context of potential budget cuts.

The *timing* attribute indicates when assistance begins: immediately within two weeks, at the start of the lean season, or at the beginning of the planting season. The immediate

option reflects the average time required for data collection, registration, and beneficiary selection. For the lean and planting season options, enumerators referred to the upcoming relevant season at the time of the interview, given that the region has two agricultural cycles per year.

The *scope* attribute varies whether assistance consists of a transfer only or is bundled with training or improved seeds. In the experimental design, bundled components are paired with lower transfer amounts, allowing us to measure beneficiaries’ willingness to trade liquidity for complementary interventions linked to graduation style programs.

The *modality* attribute distinguishes between in kind assistance, cash in hand, and cash via mobile money.

The *recipient* attribute varies whether assistance is assigned to a female or male household member, allowing us to examine preferences over intra-household allocation and control of resources. Given the program’s implicit gender tagging objective, this variation provides insight into whether policy priorities align with beneficiaries’ stated preferences.

The *targeting* attribute contrasts community-based and data-driven selection mechanisms, generating evidence relevant for refining future targeting approaches.

The six attributes and their levels imply 192 possible profiles. We implemented a D-optimal fractional factorial design in Ngene to generate choice sets that identify all main effects under a multinomial logit framework. The final design produced 48 choice pairs, divided into eight blocks of six tasks to limit cognitive burden (Kuhfeld, 2009). Households were randomly assigned to blocks, and each respondent completed six tasks. Additional design details and balance checks are provided in the Appendix.

The experiment was administered independently to one male and one female adult within each household, both completing the same six tasks. Respondents were not informed of the other household member’s choices. A comprehension check preceded the experiment, and visual aids were used to facilitate understanding. In Appendix Figure A1, we present the graphical representation of each attribute level, and in Appendix Figure A2, we present an

example of a choice set shown to respondents.

4 Analytical Framework and Estimation

4.1 Random Utility Framework

We analyze stated preferences within a random utility framework and model choices as utility maximization over bundles of observable characteristics ([Lancaster, 1966](#)). In this setting, assistance packages are not valued as indivisible goods but as vectors of attributes describing transfer size, modality, timing, scope, and recipient designation.

Let respondent n face alternative i in choice situation t . The latent utility derived from that alternative is:

$$U_{nit} = V_{nit} + \varepsilon_{nit} \quad (1)$$

where V_{nit} is the systematic component of utility and ε_{nit} is a random error term that captures unobserved factors such as idiosyncratic tastes and measurement error.

Respondent n chooses alternative i if it yields the highest utility among the available options. Under the assumption that the error terms are independently and identically distributed according to a Type I extreme value distribution, the conditional probability of choosing alternative i takes the form:

$$P_{nit}(\beta_n, \gamma) = \frac{\exp(y'_{nit}\beta_n)}{\sum_{j=1}^J \exp(y'_{njt}\beta_n)} \quad (2)$$

We estimate a mixed logit model that allows preference heterogeneity across individuals ([Hole, 2008](#); [McFadden and Train, 2000](#)), with $\beta_n \sim f(\beta \mid \theta)$. Non-transfer attributes are specified as random coefficients, while the transfer-value coefficients are treated as fixed to ensure stable willingness-to-pay estimates. All attributes are dummy coded, with the lowest level (full transfer; see below) or the reference category (for non-transfer attributes) omitted. The transfer-value attribute is encoded as three dummies for the 20%, 40%, and 60%

reductions relative to the full program amount, which serves as the omitted reference. The reference package therefore corresponds to the full transfer delivered immediately, provided in kind, without complementary components, assigned to a male recipient, and allocated through a data-driven mechanism. Coefficients are interpreted relative to this baseline. The model is estimated by simulated maximum likelihood.

Each respondent completed six choice tasks, yielding 19,332 choice situations (38,664 alternative-level observations). We first estimate the model on the pooled sample of 3,222 respondents and then separately by gender.

Willingness to pay for an attribute k is computed as the marginal rate of substitution between that attribute and the transfer amount:

$$WTP(x) = -\frac{\partial U/\partial x}{\partial U/\partial T} = -\frac{\beta_x}{\beta_T} \quad (3)$$

where β_x denotes the estimated coefficient on attribute x , and β_T is the coefficient on the transfer value. This measure represents the share of the monthly transfer respondents are willing to forgo to obtain a given program feature. Standard errors are derived using the delta method.

We selected this specification through a model comparison exercise reported in Appendix Table A10. We compared four alternatives: a conditional logit benchmark with no preference heterogeneity (M1); a mixed logit with random coefficients on all attributes including the transfer dummies (M2); our preferred mixed logit which treats the transfer-value coefficients as fixed and only the non-transfer attributes as random (M3); and a specification entering transfer value as a continuous linear and quadratic term (M4). The mixed-logit specifications dominate the conditional logit on AIC, BIC, and a likelihood-ratio test, confirming meaningful preference heterogeneity. Among the mixed-logit specifications, M3 achieves the lowest AIC and BIC and is therefore used as the main specification. The discussion of the non-linearity test in M4 is reported in the robustness section.

4.2 Gendered Constraints Within Households

While the baseline model treats respondents as independent decision makers, program attributes may be valued differently by men and women within the same household. Let $g \in \{m, f\}$ index gender. Utility can be written as:

$$U_{ghit} = \beta_g^T T_{it} + \beta_g^X X_{nit} + \gamma'_g Z_{nh} + \varepsilon_{ghit} \quad (4)$$

where Z_{nh} includes individual and household characteristics such as education, socioeconomic status, exposure to shocks, and other constraints that may shape attribute valuation.

Gender specific coefficients arise because preferences over program attributes are shaped by distinct constraint environments faced by men and women, as reflected in β_g^T and β_g^X . Attributes such as recipient designation or delivery modality reflect underlying differences in **control over resources** within the household. When transfers are assigned to one's own gender, effective control over income may increase, shaping both the marginal utility of the transfer β_g^T and the valuation of related attributes β_g^X . Conditionality and bundled components embody **time and compliance costs** that differ across gender due to unequal labor and care responsibilities, while prevailing norms and concerns about stigma further shape how program features are valued.

Socioeconomic conditions captured in γ_g , including food insecurity and education level, influence both the marginal utility of income and the perceived costs of program requirements. Given structural barriers in labor markets ([Jayachandran, 2021](#)) and financial markets ([Demirgüç-Kunt et al., 2013](#)), together with differential exposure to conflict ([Buvinić et al., 2013](#)) and climate shocks ([Afridi et al., 2022](#)), liquidity constraints, risk exposure, and **opportunity costs** may vary systematically by gender. The interaction of these structural constraints and shock exposure may therefore amplify gender differences in the valuation of assistance attributes.

We operationalize this framework by estimating the model separately by gender and

across subsamples defined by food security, household land tenure, exposure to climate and conflict shocks, women’s agency, perceptions of gender norms, and time use. Food security is measured using the Food Consumption Score (FCS) (WFP, 2008). Shock exposure captures self reported climate shocks, such as droughts or floods, and conflict related shocks experienced by the household in the previous 12 months. Gender related constraints are captured through indices of women’s decision making agency and perceived community norms regarding household decisions, constructed following approaches used in the Demographic and Health Surveys (DHS) and aggregated using the method proposed by Anderson (2008). Time use indicators are derived from a recall module of activities undertaken during the previous working day, following the American Time Use Survey methodology (ATUS, 2023). For the heterogeneity analysis, indices and continuous variables are converted into a binary indicator equal to one for observations in the top quartile of the relevant distribution. Appendix Table A2 provides variable definitions and construction details.

4.3 Intra-household differences

Assistance decisions affect multiple members of a household. When male and female respondents face identical choice scenarios yet select different alternatives, this reflects systematic differences in the valuation of program attributes across household members. Formally, if utility for gender $g \in \{m, f\}$ in household h is given by:

$$U_{ghit} = \beta_g^T T_{it} + \beta'_g X_{nit} + \varepsilon_{ghit}, \quad (5)$$

then disagreement arises whenever $\beta_m \neq \beta_f$ for at least one attribute. In this case, identical assistance packages may generate different utility maximizing choices within the same household.

We measure both the occurrence and the intensity of intra-household divergence. At the household level, we construct (i) a binary indicator equal to one if respondents differ

in at least one of the six tasks, and (ii) a continuous measure equal to the share of tasks in which selections diverge. To assess whether disagreement is concentrated along specific design dimensions, (iii) we classify alternatives into three stylized package types: liquidity-oriented packages (immediate cash delivery), investment-oriented packages (bundled with training or seeds and aligned with the planting season), and self-targeted packages (recipient matches respondent gender). For each category, we compute the share of choices in which respondents select that package type, conditional on it being available in the choice set, and construct within-household gender gaps (female selection share minus male selection share). We then estimate household-level regressions of the form:

$$Y_h = \alpha + X_h' \beta + \varepsilon_h, \quad (6)$$

where Y_h denotes disagreement measures and X_h includes socioeconomic characteristics, exposure to shocks, and measures of norms and agency. Robust standard errors are reported.

The determinants explored in the intra-household analysis also draw on the variables collected in the WFP survey. These include the variables used in the heterogeneity analysis, as well as a broader set of individual respondent characteristics and measures capturing community dynamics, including indices of social cohesion, trust, and collective action. The construction and definitions of these variables are summarized in Appendix Table [A2](#).

5 Results

5.1 Preferences for assistance attributes

Table [3](#) reports coefficients for each attribute level relative to baseline categories (full transfer amount, immediate assistance, assistance-only framing, in-kind modality, male recipient, and data-driven targeting). Coefficients therefore capture marginal utility differences with respect to these reference levels. Positive estimates indicate that a given attribute level

increases indirect utility relative to the baseline, while negative estimates imply welfare losses. Reported standard errors assess statistical precision, and the estimated standard deviations reflect unobserved preference heterogeneity across respondents in the mixed logit specification.

We begin with the transfer-value attribute. As expected, beneficiaries derive lower utility from packages that involve a cut in the monthly transfer, with the loss increasing in the magnitude of the cut. A 20% reduction is not statistically distinguishable from the full transfer in the pooled sample. A 40% reduction is associated with a marginally significant utility loss ($\beta = -0.12$, $p < 0.10$), and a 60% reduction with a significant loss ($\beta = -0.10$, $p < 0.05$). The gender-disaggregated estimates show that female respondents are substantially more sensitive to deeper cuts: a 40% reduction yields $\beta = -0.27$ ($p < 0.05$) and a 60% reduction yields $\beta = -0.21$ ($p < 0.01$) for women, whereas the corresponding male estimates are statistically indistinguishable from zero. This pattern is consistent with women bearing a larger share of the household’s consumption-smoothing burden and therefore valuing transfer continuity more strongly than men.

Turning to timing, male respondents exhibit a statistically significant preference for immediate assistance and a marginally significant aversion to transfers delivered during the planting season ($p < 0.10$). In contrast, female respondents derive higher utility from transfers delivered during the lean season relative to immediate delivery ($p < 0.10$). This pattern indicates that transfers are not primarily viewed as a vehicle for productive investment. Instead, men appear to value liquidity and immediacy, whereas women place relatively greater weight on consumption smoothing during predictable periods of seasonal hardship.

Turning to modality, both male and female respondents strictly prefer cash transfers to in-kind support ($p < 0.01$). The marginal utility of cash in hand exceeds that of mobile money, with the differential particularly pronounced among female respondents. This pattern is consistent with stronger liquidity or control considerations among women, although we do not directly observe mechanisms.

Allocation within the household reveals additional gender differences. Female respondents experience a sizable and statistically significant increase in utility when the transfer is assigned to them rather than to a male household member ($p < 0.01$). By contrast, male respondents are statistically indifferent to recipient gender. Finally, both groups express strong negative preferences for community-based targeting relative to data-driven allocation ($p < 0.01$). The magnitude and precision of these estimates indicate a clear aversion to local committee discretion, suggesting that respondents may prefer impersonal allocation mechanisms that reduce potential social frictions or favoritism (Iref).

The results indicate that beneficiaries do not view assistance primarily as a vehicle for productive investment, but rather as support aimed at meeting immediate needs and managing liquidity constraints. While preference patterns are broadly similar across gender, the strength of these preferences varies across program attributes. Men place greater weight on immediacy and liquidity, while women attach relatively greater value to timing aligned with seasonal hardship, consistent with stronger consumption smoothing concerns. Both groups prefer cash over in-kind assistance, with women showing a particularly strong preference for cash in hand. Women also derive significant additional utility when designated as the transfer recipient, whereas men are indifferent to recipient gender. Finally, the strong aversion to community-based targeting suggests a preference for impersonal allocation mechanisms, possibly reflecting concerns about discretion or local capture. Program design features beyond transfer size therefore shape welfare in meaningful and gender differentiated ways.

5.2 Willingness to pay

Appendix Table [A4](#) reports willingness-to-pay (WTP) estimates for each package attribute. As the experimental design varies the transfer amount rather than a monetary price, WTP captures the amount of the transfer respondents are willing to forgo to obtain a preferred program feature relative to its reference category. Estimates, therefore, represent implicit trade-offs in transfer value.

Table 3: Preferences for assistance attributes — simulated random parameters

	Pooled		Male		Female	
	Coef. (1)	SD (2)	Coef. (3)	SD (4)	Coef. (5)	SD (6)
Transfer value (<i>ref.: full transfer, 70 USD p.m.</i>)						
20% less	0.03 (0.05)		0.02 (0.09)		-0.04 (0.09)	
40% less	-0.12* (0.07)		-0.14 (0.11)		-0.27** (0.11)	
60% less	-0.10** (0.04)		-0.10 (0.07)		-0.21*** (0.07)	
Timing (<i>ref.: now, within two weeks</i>)						
Lean season	0.06** (0.02)	0.06 (0.08)	0.08** (0.04)	0.01 (0.04)	0.09** (0.04)	-0.04 (0.03)
Planting season	-0.04* (0.03)	0.25*** (0.06)	-0.06 (0.04)	0.41*** (0.10)	-0.02 (0.04)	-0.28* (0.15)
Scope (<i>ref.: assistance only</i>)						
Plus improved seeds	0.12*** (0.03)	0.00 (0.02)	0.14*** (0.04)	-0.16 (0.12)	0.17*** (0.04)	-0.16 (0.15)
Plus training	0.28*** (0.03)	0.24*** (0.07)	0.31*** (0.05)	0.37*** (0.12)	0.35*** (0.05)	-0.36*** (0.13)
Modality (<i>ref.: in-kind</i>)						
Cash in hand	0.35*** (0.03)	0.35*** (0.05)	0.40*** (0.04)	0.64*** (0.07)	0.47*** (0.05)	0.71*** (0.06)
Mobile money	0.15*** (0.03)	0.26*** (0.06)	0.20*** (0.05)	0.56*** (0.09)	0.19*** (0.04)	-0.44*** (0.12)
Recipient: female	0.10*** (0.02)	0.27*** (0.04)	-0.02 (0.03)	0.59*** (0.06)	0.24*** (0.03)	0.54*** (0.06)
Targeting: community-based	-0.63*** (0.02)	0.63*** (0.03)	-0.78*** (0.05)	1.09*** (0.06)	-0.73*** (0.04)	1.04*** (0.06)
Observations	38,460		19,128		19,332	
No. respondents	1,611		1,594		1,611	
LR chi2	921.8		374.0		415.5	
Log-likelihood	-12228		-5969		-6022	
Halton draws	100		100		100	

Coefficients are interpreted relative to the indicated reference category. SD columns report the standard deviation of the random-coefficient distribution and capture preference heterogeneity; cells are blank for transfer-value rows because those coefficients are fixed by construction (see Section 4). Standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. Observations reflect the six choice tasks per respondent.

Complementary program components receive the highest valuations. Male respondents are willing to forgo 174,000 CDF ($\approx$ USD 76) to receive training and 75,400 CDF ($\approx$ USD 33) to receive improved seeds. Female respondents display a similar ranking, though with smaller magnitudes, valuing training at 118,400 CDF ($\approx$ USD 52) and seeds at 49,100 CDF ($\approx$ USD 21). Delivery modality is also valued. Both male and female respondents place a higher valuation on receiving assistance in cash relative to the reference modality. Male respondents are willing to forgo 198,700 CDF ($\approx$ USD 87) to receive cash in hand and 92,600 CDF ($\approx$ USD 41) for transfers delivered via mobile money. Female respondents rank similarly, with corresponding valuations of 154,200 CDF ($\approx$ USD 67) and 54,900 CDF ($\approx$ USD 24). The larger valuation of cash in hand suggests that immediate liquidity is valued more strongly than digital transfers.

Valuations differ more markedly with respect to the designated recipient of assistance. Male respondents show no statistically significant valuation of the recipient’s gender, indicating indifference about which household member receives the transfer. Female respondents, however, are willing to forgo 80,500 CDF ($\approx$ USD 35) to direct transfers to women, suggesting that women place value on female receipt of assistance.

Finally, respondents place substantial value on data-driven targeting relative to community-based targeting. Male respondents are willing to forgo 390,300 CDF ($\approx$ USD 171) to receive data-driven targeting, while the corresponding valuation among female respondents is 232,800 CDF ($\approx$ USD 102).

5.3 Heterogeneities

We present heterogeneity in preferences for each respondent across the potential characteristics described above, where differences in preferences may be expected.

Male respondents

Appendix Tables [A5](#) and [A6](#) show the WTP heterogeneity estimates for the male sample. Across attributes, heterogeneity appears more pronounced along gender related dimensions,

particularly female agency and reproductive labor gaps, as well as exposure to shocks, while it is more limited across vulnerability indicators such as food insecurity. Preferences for transfer modality and productive assistance bundles remain relatively stable across most groups, with only modest differences in the magnitude of willingness to pay, whereas timing attributes exhibit more context specific variation. Specifically:

Food insecurity. Food secure households would be willing to forgo about 77% of the transfer to receive assistance via mobile money, compared with 39% among food insecure households, while food insecure households value cash in hand about 10% more, indicating a stronger preference for direct cash transfers.

Land ownership. Male respondents in households that own or rent land display clear and statistically significant preferences for assistance bundled with seeds and training, for cash in hand and mobile money, and for data driven targeting, alongside a marginal preference for transfers delivered during the lean season. In contrast, male respondents in landless households show no statistically significant preferences across attributes, with estimates that are imprecise relative to the reference package. This pattern suggests that male preferences over assistance design are more clearly articulated when households have a productive base.

Exposure to climate shocks. Males living in households affected by climate shocks display negative preferences for transfers delivered during the planting season but show strong preferences for training and cash in hand relative to their reference categories.

Exposure to conflict. Male respondents in households affected by conflict appear largely indifferent to the package component, while those not affected show strong preferences for immediate assistance, training, and cash in hand.

Women's agency. In households with low levels of female agency, both within the household and in community perceptions, respondents show strong preferences for training and cash in hand. In contrast, in households with higher levels of female agency, preferences are largely indifferent relative to the reference package, with the exception of training, which remains preferred over assistance alone or assistance with seeds.

Female work burden. In households where women work fewer hours, respondents show strong preferences for assistance bundled with seeds, training, and both cash modalities, particularly cash in hand. In contrast, in households where women work substantial hours, preferences remain positive but are generally smaller in magnitude, and only training and cash in hand remain statistically preferred relative to their reference categories.

Reproductive labor gap. Households with larger reproductive labor gaps show slightly stronger preferences for seeds but weaker preferences for mobile money.

Climate shocks and food insecurity. Households experiencing both climate shocks and food insecurity show weaker preferences for mobile money and seeds and a stronger negative preference for planting season transfers compared with households not jointly exposed to these conditions.

Conflict exposure and food insecurity. Households jointly affected by conflict and food insecurity also display weaker preferences for additional program features compared with those not exposed to both conditions. In particular, preferences for mobile money and other complementary components are less pronounced, indicating a tendency to remain closer to the default assistance package.

Male respondents in households facing higher levels of vulnerability or exposure to shocks do not systematically express stronger preferences for additional liquidity or investment oriented features. Instead, their preferences tend to remain closer to the default assistance package, with relatively greater value placed on immediacy and more traditional forms of support. In contrast, heterogeneity in preferences is more pronounced along gender related dimensions, where differences in female agency, time burdens, and intra household labor allocation are associated with stronger shifts in the relative valuation of program features such as training and transfer modality.

Female respondents

Appendix Tables [A7](#) and [A8](#) present the WTP heterogeneity estimates for the female sample. Preference patterns remain broadly stable across groups, with complementary com-

ponents and cash modalities consistently valued. Heterogeneity is more pronounced along gender related dimensions, particularly female agency and reproductive labor burdens, as well as exposure to shocks, while differences across standard vulnerability indicators such as food insecurity are more limited. More specifically:

Food insecurity. Female respondents from food secure households show a preference for transfers delivered during the lean season, while food insecure households appear to prefer more immediate assistance. Food insecure households also display a stronger preference for cash in hand, whereas food secure households show relatively stronger preferences for being the transfer recipient.

Land ownership. Female respondents in both land owning and landless households display the same qualitative preference structure, with positive and statistically significant valuations for assistance bundled with seeds and training, for both cash modalities, for being designated as the recipient, and for data driven targeting. Magnitudes, however, are systematically larger among land owning households, particularly for training, cash in hand, and the female recipient designation, suggesting that women in households with a productive base place greater weight on complementary components and on directing assistance to themselves than women in landless households.

Exposure to climate shocks. Similar to male respondents, female respondents in households affected by climate shocks show strong preferences for training over seeds, as well as for cash in hand and self targeted assistance, whereas households not affected by climate shocks display no clear preferences relative to the reference package.

Exposure to conflict. Unlike male respondents in households affected by conflict, who show no differences in preferences relative to the reference package, female respondents display clear preferences for alternatives such as seeds, training, and cash in hand. A similar pattern is observed among female respondents in households not affected by conflict, with the additional preference for mobile money over the in kind option.

Women's agency. Female respondents in households with lower female agency show

stronger preferences for seeds, training, and cash in hand, whereas among households with higher female agency, seeds are no longer preferred relative to the reference package and preferences are generally smaller in magnitude. When community perceptions of women's agency are low, meaning women are not expected to make decisions over resources, the main difference relative to male respondents relates to the timing of transfers. Male respondents in low agency households prefer transfers delivered during the lean season, whereas female respondents are largely indifferent to timing and show a negative preference for transfers delivered during the planting season.

Female work burden. Female respondents with longer working hours display preference patterns similar to those of women with lower time burdens across most attributes, with the exception of mobile money, which is less preferred among the latter. This contrasts with men, who exhibit greater differences, particularly in their valuation of complementary program components and of receiving transfers via mobile money, which are more preferred by men living in households where women have lower working hours.

Reproductive labor gap. Preference patterns remain largely stable across households with larger and smaller gender gaps in reproductive labor for both male and female respondents. Differences mainly appear in the magnitude and precision of some estimates, particularly for mobile money, which becomes less preferred when the reproductive labor gap is larger.

Climate shocks and food insecurity. Preference patterns remain broadly similar across households with and without joint exposure to climate shocks and food insecurity. Differences mainly appear in the magnitude of some attributes: assistance during the planting season becomes more strongly disliked among affected households, while complementary components and cash modalities remain strongly valued. Preferences for mobile money, however, weaken under compounded vulnerability.

Conflict exposure and food insecurity. Preference patterns are similar to those of male respondents for most attributes, except for targeting. While male respondents do not significantly value data driven targeting when exposed to both conflict and food insecurity, it

remains the most preferred method among female respondents in these households.

Female respondents display somewhat greater variation in preferences across household characteristics, although core valuations remain broadly stable. Timing preferences vary with household conditions, with food secure households preferring transfers delivered during the lean season, while food insecure households place greater value on more immediate assistance. Exposure to climate shocks is associated with clearer preferences for complementary components such as training and for cash modalities, whereas households not exposed to shocks remain closer to the reference assistance package.

Comparing these patterns with those observed among male respondents reveals broadly similar preference structures across most attributes. However, several differences emerge. Female respondents tend to express clearer and more consistent preferences under conditions of vulnerability, particularly among households affected by conflict or compounded shocks, where complementary components and cash modalities remain strongly valued. In contrast, male respondents in similar contexts often display weaker or statistically insignificant preferences relative to the reference package. Differences also arise in specific attributes. For example, mobile money tends to lose salience among men under compounded climate and food insecurity shocks, while women maintain clearer preferences for transfer modalities. Similarly, in households experiencing both conflict and food insecurity, women display a strong preference for data driven targeting, whereas male respondents show no significant valuation for this attribute. Overall, while core preferences are largely shared across genders, women’s valuations appear more responsive to gender related constraints and vulnerability contexts.

5.4 Intra-household differences

Our data suggests meaningful differences in decision making within households. On average, partners choose different assistance packages in roughly one third of the choice scenarios, corresponding to about two decisions across the experiment. Disagreement is also widespread:

around 74% of households exhibit at least one instance in which partners select different options.

When examining the type of assistance package chosen, preferences appear broadly similar for liquidity oriented and investment oriented packages. Approximately 55% of both male and female respondents choose packages providing immediate liquidity when these are available, while around 52% select investment oriented packages when presented with them. Differences emerge more clearly for packages that target the respondent’s own gender. Women are about 3 percentage points more likely ($p < 0.01$) to choose packages in which the transfer is directed to them than male respondents are to choose packages directed to men.

Appendix Table A9 examines the determinants of intra household disagreement, considering both the likelihood that partners disagree at least once (column 1) and the overall disagreement rate across choice tasks (column 2). Households reporting stronger social cohesion exhibit significantly lower levels of disagreement. A one standard deviation increase in the social cohesion index is associated with a lower probability of any disagreement ($p < 0.01$) and a lower disagreement rate ($p < 0.01$). Similarly, higher levels of trust within the community are linked to lower disagreement rates ($p < 0.10$).

Greater male engagement in work and care activities is also associated with more aligned preferences over assistance packages. More hours worked by men reduce the likelihood of disagreement ($p < 0.01$), and greater male involvement in care work is associated with both a lower probability of disagreement ($p < 0.05$) and a lower disagreement rate ($p < 0.05$). Higher male depression scores are likewise associated with lower disagreement, both in the probability of any disagreement ($p < 0.05$) and in the disagreement rate ($p < 0.10$).

Indicators of vulnerability show mixed associations. Households that report severe coping strategies are more likely to exhibit at least one disagreement ($p < 0.01$). By contrast, households with higher women’s diet diversity ($p < 0.01$) and those classified as food insecure ($p < 0.01$) show lower probabilities of disagreement. Larger gender gaps in care work are

also associated with a lower likelihood of disagreement ($p < 0.01$). Households where the male respondent belongs to the Twa ethnic group ($p < 0.10$) and those where the female respondent is internally displaced ($p < 0.10$) also exhibit lower probabilities of disagreement.

Exposure to conflict is associated with higher disagreement rates ($p < 0.05$), and when combined with low male education it increases the likelihood of any disagreement ($p < 0.10$), while climate shocks are associated with lower disagreement rates ($p < 0.01$). Gender norms also appear relevant: stronger female agency attitudes among women are associated with higher disagreement rates ($p < 0.05$). By contrast, greater male support for female agency and more supportive partner attitudes are associated with lower disagreement, although these effects are not statistically significant. Finally, households with larger care gaps ($p < 0.05$) and greater female work hours ($p < 0.10$) exhibit lower disagreement rates.

Column 3 in Appendix Table [A9](#) examines the determinants of the intra household gap in the probability of choosing packages that target the respondent’s own gender, which is found to be larger among female respondents. Higher levels of trust within the community are associated with women being more likely than men to prefer packages directed to them ($p < 0.10$). Similarly, households that own or rent land are associated with women being more likely than men to prefer packages directed to female recipients ($p < 0.10$). In contrast, higher levels of female life satisfaction are associated with smaller gaps ($p < 0.05$), indicating that households where women report higher subjective wellbeing exhibit more aligned preferences regarding the gender of the recipient. Community norms more supportive of female agency ($p < 0.01$) and higher levels of female life satisfaction ($p < 0.05$) are associated with smaller gender gaps in the selection of female recipients, indicating that men and women move closer in their preferences regarding whether assistance should be directed to women.

5.5 Robustness

We assess the robustness of the preferred specification along three dimensions. First, we test the independence-of-irrelevant-alternatives (IIA) assumption underlying the conditional logit

using a Hausman test that compares the full conditional logit with a deducted specification dropping the targeting attribute (Appendix Table A3). Although our preferred specification is a mixed logit, which relaxes IIA by construction, the test offers a conventional check on the simpler benchmark.

Second, we compare four alternative specifications of the choice model on log-likelihood, AIC, and BIC (Appendix Table A10). The conditional logit benchmark (M1) is rejected against the mixed-logit alternatives by a likelihood-ratio test ($\chi^2(11) = 384.4$, $p < 0.001$), confirming meaningful preference heterogeneity across respondents. Among the mixed-logit specifications, the model that treats the transfer-value coefficients as fixed and only the non-transfer attributes as random (M3) achieves the lowest AIC (24,517) and BIC (24,680), supporting our choice of this specification for the main results. A specification entering transfer value as a continuous linear and quadratic term (M4) yields a slightly worse fit, suggesting that the dummy encoding adequately captures the response surface within the cut range studied.

Third, we test for attribute non-attendance (ANA), a common concern in discrete choice experiments where respondents may simplify the task by ignoring one or more attributes (Hensher et al., 2005). We follow a constrained-model approach: for each of the six attribute groups we re-estimate the mixed logit with the corresponding mean and standard-deviation coefficients restricted to zero, and compare the constrained log-likelihood with the unrestricted model using a likelihood-ratio test (Appendix Table A11). The null of non-attendance is rejected at the 5% level or better for every attribute, with likelihood-ratio statistics ranging from 11.3 ($p = 0.023$) for timing to 1,610.1 ($p < 0.001$) for the targeting mechanism. The very large statistic on targeting reinforces the headline finding that respondents have particularly strong preferences over how beneficiaries are selected, and the pattern overall confirms that respondents engaged with every design dimension. This provides additional confidence that the willingness-to-pay estimates reflect substantive trade-offs across the full set of attributes rather than partial attribute attendance.

6 Conclusion and Policy Implications

Beneficiaries assess humanitarian assistance not only by its monetary value but also by features such as timing, modality, recipient designation, and the rule used to select participants. Our discrete choice experiment with 3,222 matched male and female respondents identifies several patterns. Beneficiaries treat assistance as liquidity support rather than as productive investment: cash dominates in-kind delivery for both genders, and timing aligned with immediate consumption (men) or with the lean season (women) is preferred over disbursements scheduled to the planting season. Gender shapes preferences asymmetrically. Women place a premium of about USD 35 on being designated as the recipient and value cash-in-hand more strongly than men, while men are statistically indifferent to recipient identity; the heterogeneity gradient tracks female agency, time burdens, and the within-household care gap more closely than food insecurity or shock exposure. The single largest effect concerns the allocation rule: respondents forgo upwards of one full monthly transfer to be selected through a data-driven mechanism rather than by a community committee. Disagreement within the household is widespread but not random—partners diverge on roughly one in three choice tasks, with disagreement falling in households with stronger social cohesion, higher trust, and greater male participation in work and care, rather than with material deprivation.

These patterns have direct bearing on program design under tightening humanitarian budgets. Two ostensibly cost-neutral changes—shifting cash-in-hand to mobile money or rescheduling disbursements to the planting season—impose welfare losses comparable in magnitude to a 40% cut in the nominal transfer. Female-recipient designation generates additional welfare for women, but it operates through perceived control rather than mechanical earmarking, and is therefore unlikely to produce equivalent gains in households where women lack agency. The aversion to community-based targeting suggests that the perceived fairness of the allocation rule shapes program valuation independently of its accuracy, and that the welfare costs of discretionary selection may be larger than commonly assumed.

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Appendix

Figure A1: Graphical presentation of all assistance attributes

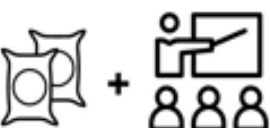
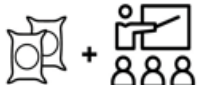
Forfait d'assistance	A	B
La valeur de l'assistance:	 70,000 CDF pm	 160,000 CDF pm
à quelle époque:	 Pendant la période de plantation	 Pendant la saison creuse
Quel forfait d'assistance:	 Assistance et formation	 Assistance seulement
Comment le recevoir:	 Argent liquide en main	 Transfert en nature (valeur équivalente)
Qui le reçoit:	 Homme	 Femme
Comment sélectionner bénéficiaires:	 Basé sur des données/enquêtes	 Processus d'autosélection communautaire

Figure A2: Example of a choice set presented to beneficiaries within the survey

Forfait d'assistance				
La valeur de l'assistance:	 160,000 CDF pm	 130,000 CDF pm	 100,000 CDF pm	 70,000 CDF pm
à quelle époque:	 Dans deux semaine	 Pendant la saison creuse	 Pendant la période de plantation	
Quel forfait d'assistance:	 Assistance seulement	 Assistance et formation	 Assistance et semences	
Comment le recevoir:	 Argent liquide en main	 L'argent mobile (sur le telephone)	 Transfert en nature (valeur équivalente)	
Qui le reçoit:	 Femme	 Homme		

Appendix Tables

Table A1: Number of sample villages and households

Village grouping	Number of villages	Number of households
Walungu		
Izege	14	105
Kaniola	22	293
Mulamba	19	279
Mwenga		
Balobola	6	94
Bamulinda	8	163
Bashilubanda	10	160
Bashimwenda 1Er	6	82
Batumba	3	70
Irangi	4	108
Kigogo	10	146
Kyonvu	4	111
Total	106	1,611

Notes: Sample villages and households across the two territories of South Kivu (Mwenga and Walungu) included in the WFP–DIME 2023 fieldwork. Villages are grouped by the farmer organisation cluster (*groupement*) used by the Joint Resilience Programme. Households were selected proportionally to village population size using a random-walk procedure; both the male and female household heads were interviewed when available. *Source:* Authors' construction based on the WFP Gender, Social Cohesion, and Resilience survey, South Kivu, 2023.

Table A2: Details on variable construction

Indicator	Definition / construction
Food security (FCS)	Food security is measured using the Food Consumption Score (FCS), an indicator commonly used by WFP that aggregates the frequency of consumption of different food groups over the previous seven days, weighted by their nutritional value (WFP, 2008). A binary indicator is constructed equal to one for households in the top quartile of the distribution.
Land ownership	Binary indicator equal to one if the household reports owning or renting agricultural land used for cultivation by household members, and zero otherwise. Constructed from the household roster module of the WFP survey.
Shock exposure	Measured using respondents' self-reported experiences of shocks affecting the household in the previous 12 months. Climate shocks include drought or irregular rainfall, floods, landslides or erosion, and hail or frost. Conflict shocks include land conflicts, militia group activity, and ethnic or religious conflicts. Binary indicators are constructed equal to one if the household reports experiencing at least one shock in each category. Additional indicators capture severe shocks, defined as households reporting at least one shock with high reported severity.
Women's agency index	Constructed following the approach used in the Demographic and Health Surveys (DHS), aggregating women's responses to questions on who should make decisions in four domains: major household purchases, purchases from men's income, purchases from women's income, and women's health care. Items are aggregated using the Anderson (2008) procedure, then converted into a binary indicator equal to one for households in the top quartile of the distribution.
Community perceptions index	Constructed from women's responses to questions on who people in their community think should usually make decisions in the same four domains as the agency index. Items are aggregated using Anderson (2008), then converted into a binary indicator equal to one for households in the top quartile.
Reproductive work time	Measured using a recall module in which respondents report activities undertaken during the previous working day and their duration, following the American Time Use Survey methodology (ATUS, 2023). Reproductive work time is the total time spent on unpaid household activities, including chores, collecting firewood and water, other domestic activities, and childcare. A binary indicator equals one for individuals in the top quartile.
Total working time	Using the same time-use module, total working time is the sum of time spent on agricultural and non-agricultural activities, including wage employment and self-employment. A binary indicator equals one for individuals in the top quartile.
Social cohesion index	Captures trust, perceived honesty of community members, perceived closeness to the community, and participation in collective activities (group membership, volunteering, community contributions). Variables are aggregated into a standardized index using Anderson (2008), with higher values indicating stronger social cohesion.
Trust index	Captures interpersonal and institutional trust, constructed from respondents' reported levels of trust in extended family, close neighbors, community members, local leaders, the national government, and strangers. Aggregated into a standardized index using Anderson (2008).
Collective action index	Measured through respondents' participation in groups, positions of responsibility within groups, volunteering, and contributions to community projects. Aggregated into a standardized index using Anderson (2008).

Source: Authors' construction based on the WFP Gender, Social Cohesion, and Socio-Economic Resilience survey modules, South Kivu (Trako et al., 2024).

Table A3: Testing for independence of irrelevant alternatives (IIA)

	Full (b)	Deducted (B)	Difference (b-B)	S.E. (b-B)
Transfer size: minus 20%	0.05	0.16	-0.12	0.01
Transfer size: minus 40%	-0.12	0.18	-0.29	0.02
Transfer size: minus 60%	-0.09	-0.02	-0.07	0.01
Timing: lean season	0.06	0.05	0.02	0.00
Timing: planting season	-0.04	-0.03	-0.01	0.01
Scope: assistance + seeds	0.10	0.05	0.06	0.01
Scope: assistance + training	0.27	0.24	0.03	0.01
Modality: cash in hand	0.31	0.28	0.03	0.01
Modality: mobile money	0.13	0.10	0.03	0.00
Recipient: female	0.09	0.06	0.03	0.00
Targeting: community-based	-0.56			
Constant	-0.20	-0.18	-0.03	0.01
<i>Observations</i>	38,460	38,460		
<i>No. respondents</i>	1,611	1,611		
<i>Log-likelihood</i>	-12354.69	-13028.59		
Hausman test:	$\chi^2(10) = -632.53, \quad p = 1.000$			

*Source: Authors' calculations on the pooled sample. Full = conditional logit with all attributes; Deducted = conditional logit excluding the targeting attribute. Hausman test compares the two specifications. *** p < 0.01, ** p < 0.05, * p < 0.1.*

Table A4: Willingness to pay for assistance attributes ('000s CDF)

	Male				Female			
	Coef.	SE	95% CI		Coef.	SE	95% CI	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Timing (<i>ref.: now, within two weeks</i>)								
Lean season	30.27	(21.23)	-11.35	71.88	17.97	(12.04)	-5.62	41.57
Planting season	-38.68	(23.53)	-84.80	7.43	-18.39	(13.39)	-44.65	7.86
Scope (<i>ref.: assistance only</i>)								
Plus improved seeds	75.41***	(24.18)	28.02	122.81	49.08***	(12.41)	24.75	73.41
Plus training	174.04***	(46.28)	83.33	264.75	118.44***	(20.67)	77.92	158.96
Modality (<i>ref.: in-kind</i>)								
Cash in hand	198.70***	(53.75)	93.35	304.04	154.24***	(26.60)	102.11	206.36
Mobile money	92.63***	(31.21)	31.46	153.80	54.88***	(15.00)	25.49	84.27
Recipient: female	-11.70	(16.56)	-44.15	20.75	80.54***	(15.17)	50.81	110.28
Targeting: data-driven	390.31***	(101.79)	190.81	589.82	232.78***	(37.39)	159.49	306.07
Observations		19,128				19,332		
No. respondents		1,594				1,611		
LR chi2		375.3				457.1		
Log-likelihood		-5961				-5935		
Halton draws		100				100		

Coefficients are in thousands of CDF and represent willingness to pay, i.e. the amount of transfer value respondents are willing to forgo for a given attribute level relative to the reference category. Estimates from mixed logit models. Standard errors in parentheses, clustered at the respondent level. Columns (3)–(4) and (7)–(8) report 95% confidence intervals. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. Each respondent contributes six choice-task observations. 100 Halton draws.

Table A5: Preferences heterogeneities — Male (vulnerability, shocks, joint exposures)

Male respondents	Food sec.	Food inse.	No land	Owns land	Climate (no)	Climate (yes)	Conflict (no)	Conflict (yes)	No clim. × food sec.	Clim. × food inse.	No conf. × food sec.	Conf. × food inse.
Lean season	0.424 (0.292)	0.184 (0.394)	0.346 (1.817)	0.394* (0.222)	13.691 (140.929)	0.193 (0.203)	0.293 (0.215)	7.744 (119.184)	0.472 (0.291)	-0.088 (0.455)	0.295 (0.229)	5.666 (69.406)
Planting season	-0.220 (0.278)	-0.694 (0.553)	-1.818 (4.237)	-0.278 (0.226)	6.284 (65.325)	-0.465* (0.239)	-0.269 (0.226)	-17.198 (265.533)	-0.183 (0.267)	-1.040 (0.805)	-0.344 (0.244)	-13.642 (169.666)
Plus seeds	0.774** (0.307)	0.868* (0.515)	0.067 (1.590)	0.919*** (0.267)	9.499 (95.692)	0.696*** (0.231)	0.713*** (0.239)	14.381 (218.382)	0.850*** (0.318)	0.552 (0.488)	0.745*** (0.256)	7.783 (93.826)
Plus training	1.750*** (0.596)	1.970** (0.995)	5.008 (10.478)	1.659*** (0.442)	7.254 (73.342)	1.762*** (0.460)	1.683*** (0.456)	32.836 (502.750)	1.569*** (0.525)	2.552* (1.478)	1.756*** (0.498)	18.973 (232.399)
Cash in hand	2.113*** (0.726)	2.266* (1.183)	6.382 (13.613)	1.860*** (0.507)	26.271 (270.121)	1.872*** (0.501)	1.998*** (0.543)	31.432 (483.289)	2.092*** (0.697)	2.396* (1.455)	2.161*** (0.617)	6.129 (76.677)
Mobile money	1.226*** (0.470)	0.621 (0.526)	2.708 (5.918)	0.834*** (0.296)	11.185 (114.797)	0.905*** (0.302)	0.863*** (0.307)	21.913 (336.089)	1.256*** (0.467)	0.117 (0.504)	1.054*** (0.364)	-3.303 (43.017)
Female recipient	-0.058 (0.214)	-0.328 (0.353)	-2.033 (4.547)	0.001 (0.159)	-10.578 (109.212)	-0.054 (0.160)	-0.113 (0.167)	-4.105 (64.106)	-0.205 (0.219)	-0.015 (0.358)	-0.043 (0.174)	-7.429 (92.056)
Data targeting	3.984*** (1.327)	4.502** (2.264)	18.855 (39.971)	3.138*** (0.824)	44.610 (459.883)	3.665*** (0.930)	3.537*** (0.938)	104.675 (1608.201)	3.948*** (1.272)	4.871* (2.855)	4.058*** (1.130)	45.270 (556.024)
<i>Observations</i>	11,604	7,524	5,964	13,164	3,696	15,432	14,832	4,296	13,404	5,724	17,220	1,908

Notes: Each column reports willingness-to-pay (WTP) estimates from a mixed logit model fitted on the subsample defined by the column header (heterogeneity dimension and binary level). Coefficients are in 100,000 CDF units (multiply by 100 for '000s CDF). Reference categories: full transfer (160,000 CDF), immediate timing, in-kind modality, male recipient, data-driven targeting. Standard errors clustered at the respondent level in parentheses; computed via the delta method. Each respondent contributes six choice tasks. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table A6: Preferences heterogeneities — Male (gender-related constraints)

Male respondents	Agency <p75	Agency ≥p75	Comm. perc. <p75	Comm. perc. ≥p75	Work time <p75	Work time ≥p75	Repro gap <p75	Repro gap ≥p75
Lean season	0.516 (0.427)	0.134 (0.244)	0.448* (0.272)	-0.172 (0.526)	0.470 (0.320)	0.197 (0.340)	0.381 (0.277)	0.214 (0.454)
Planting season	-0.654 (0.481)	-0.066 (0.259)	-0.448 (0.282)	-0.110 (0.527)	-0.320 (0.307)	-0.475 (0.414)	-0.404 (0.294)	-0.437 (0.509)
Plus seeds	1.243** (0.602)	0.297 (0.226)	0.941*** (0.319)	0.178 (0.478)	0.889** (0.364)	0.519 (0.345)	0.728** (0.302)	0.968* (0.563)
Plus training	2.593** (1.201)	0.995*** (0.351)	1.555*** (0.484)	2.935 (1.981)	2.112*** (0.758)	1.021** (0.491)	1.863*** (0.617)	1.759* (0.992)
Cash in hand	3.744** (1.750)	0.413 (0.267)	2.191*** (0.678)	1.649 (1.206)	2.663*** (0.971)	0.882* (0.496)	2.275*** (0.759)	1.921* (1.150)
Mobile money	1.570* (0.801)	0.382 (0.270)	0.902** (0.362)	1.247 (0.954)	1.264** (0.526)	0.393 (0.390)	1.113** (0.434)	0.698 (0.616)
Female recipient	-0.044 (0.287)	-0.204 (0.212)	0.002 (0.189)	-0.592 (0.588)	-0.063 (0.225)	-0.325 (0.305)	-0.117 (0.209)	-0.100 (0.354)
Data targeting	6.510** (3.035)	1.709*** (0.532)	4.076*** (1.223)	4.537 (3.099)	4.507*** (1.619)	3.664** (1.523)	4.069*** (1.325)	4.488* (2.499)
<i>Observations</i>	15,108	4,020	14,412	4,716	14,664	4,392	14,412	4,716

Notes: Each column reports willingness-to-pay (WTP) estimates from a mixed logit model fitted on a subsample defined by a gender-related heterogeneity dimension. Coefficients are in 100,000 CDF units. “< p75” (“≥ p75”) denotes households below (at or above) the 75th percentile of the relevant index. Reference categories: full transfer (160,000 CDF), immediate timing, in-kind modality, male recipient, data-driven targeting. Standard errors clustered at the respondent level in parentheses; computed via the delta method. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table A7: Preferences heterogeneities — Female (vulnerability, shocks, joint exposures)

Female respondents	Food sec.	Food insec.	No land	Owns land	Climate (no)	Climate (yes)	Conflict (no)	Conflict (yes)	No clim. × food sec.	Clim. × food insec.	No conf. × food sec.	Conf. × food insec.
Lean season	0.324* (0.174)	0.123 (0.232)	0.286 (0.199)	0.221 (0.191)	0.792 (0.742)	0.174 (0.144)	0.262 (0.172)	0.216 (0.243)	0.273* (0.159)	0.139 (0.314)	0.286* (0.154)	0.045 (0.314)
Planting season	0.112 (0.179)	-0.500* (0.287)	0.152 (0.217)	-0.291 (0.206)	0.872 (0.860)	-0.230 (0.157)	-0.128 (0.181)	-0.086 (0.256)	0.159 (0.167)	-1.039** (0.498)	-0.027 (0.158)	-0.638 (0.480)
Plus seeds	0.637*** (0.186)	0.499** (0.235)	0.380** (0.193)	0.699*** (0.214)	1.076 (0.801)	0.531*** (0.147)	0.586*** (0.174)	0.624** (0.294)	0.629*** (0.173)	0.490 (0.299)	0.645*** (0.163)	0.144 (0.335)
Plus training	1.239*** (0.293)	1.209*** (0.407)	0.846*** (0.283)	1.475*** (0.370)	1.437 (1.106)	1.227*** (0.247)	1.307*** (0.295)	1.171** (0.463)	1.178*** (0.266)	1.632** (0.655)	1.343*** (0.276)	0.339 (0.414)
Cash in hand	1.512*** (0.366)	2.043*** (0.621)	1.336*** (0.400)	1.918*** (0.482)	2.471 (1.766)	1.680*** (0.335)	2.039*** (0.446)	0.856** (0.397)	1.569*** (0.352)	2.316** (0.921)	1.843*** (0.371)	0.769 (0.544)
Mobile money	0.663*** (0.222)	0.544* (0.295)	0.595** (0.260)	0.639*** (0.241)	0.960 (0.856)	0.592*** (0.182)	0.718*** (0.228)	0.407 (0.288)	0.698*** (0.211)	0.399 (0.369)	0.658*** (0.195)	0.314 (0.426)
Female recipient	0.901*** (0.228)	0.669*** (0.257)	0.636*** (0.219)	1.010*** (0.274)	1.461 (1.151)	0.803*** (0.174)	0.894*** (0.220)	0.820** (0.329)	0.924*** (0.218)	0.719** (0.347)	0.927*** (0.203)	0.498 (0.336)
Data targeting	2.475*** (0.556)	2.848*** (0.850)	2.542*** (0.680)	2.608*** (0.631)	3.182 (2.294)	2.568*** (0.478)	2.742*** (0.590)	2.314*** (0.799)	2.376*** (0.497)	3.506*** (1.352)	2.682*** (0.520)	2.104** (1.005)
<i>Observations</i>	11,724	7,608	6,048	13,284	3,744	15,588	14,976	4,356	13,560	5,772	17,412	1,920

Notes: Each column reports willingness-to-pay (WTP) estimates from a mixed logit model fitted on the subsample defined by the column header (heterogeneity dimension and binary level). Coefficients are in 100,000 CDF units (multiply by 100 for '000s CDF). Reference categories: full transfer (160,000 CDF), immediate timing, in-kind modality, male recipient, data-driven targeting. Standard errors clustered at the respondent level in parentheses; computed via the delta method. Each respondent contributes six choice tasks. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table A8: Preferences heterogeneities — Female (gender-related constraints)

Female respondents	Agency <p75	Agency ≥p75	Comm. perc. <p75	Comm. perc. ≥p75	Work time <p75	Work time ≥p75	Repro gap <p75	Repro gap ≥p75
Lean season	0.404** (0.181)	-0.207 (0.240)	0.179 (0.131)	0.649 (0.797)	0.166 (0.166)	0.450* (0.261)	0.320** (0.163)	0.025 (0.295)
Planting season	-0.088 (0.177)	-0.292 (0.270)	-0.281* (0.146)	1.327 (1.335)	-0.105 (0.177)	-0.132 (0.257)	-0.053 (0.169)	-0.388 (0.334)
Plus seeds	0.701*** (0.188)	0.287 (0.238)	0.538*** (0.135)	0.953 (0.906)	0.625*** (0.180)	0.445* (0.239)	0.616*** (0.176)	0.508* (0.276)
Plus training	1.462*** (0.325)	0.709** (0.329)	1.006*** (0.193)	3.098 (2.681)	1.437*** (0.326)	0.776** (0.319)	1.185*** (0.270)	1.445*** (0.559)
Cash in hand	1.905*** (0.423)	1.092** (0.428)	1.454*** (0.269)	3.371 (2.910)	1.865*** (0.421)	1.213*** (0.433)	1.743*** (0.383)	1.627** (0.670)
Mobile money	0.628*** (0.216)	0.664** (0.318)	0.554*** (0.162)	0.953 (1.032)	0.685*** (0.221)	0.399 (0.276)	0.746*** (0.224)	0.383 (0.323)
Female recipient	0.841*** (0.211)	0.890*** (0.337)	0.787*** (0.156)	1.299 (1.226)	0.894*** (0.221)	0.831*** (0.310)	0.812*** (0.202)	0.963** (0.397)
Data targeting	2.851*** (0.605)	1.912*** (0.656)	2.087*** (0.358)	6.128 (5.291)	2.697*** (0.578)	2.046*** (0.660)	2.432*** (0.514)	3.298*** (1.196)
<i>Observations</i>	15,252	4,080	14,568	4,764	14,784	4,476	14,580	4,752

Notes: Each column reports willingness-to-pay (WTP) estimates from a mixed logit model fitted on a subsample defined by a gender-related heterogeneity dimension. Coefficients are in 100,000 CDF units. “< p75” (“≥ p75”) denotes households below (at or above) the 75th percentile of the relevant index. Reference categories: full transfer (160,000 CDF), immediate timing, in-kind modality, male recipient, data-driven targeting. Standard errors clustered at the respondent level in parentheses; computed via the delta method. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table A9: Intra-household differences in choice — determinants

	Any disagree. (1)	Disagree. rate (2)	Gap: liquidity (3)	Gap: own gender (4)	Gap: investment (5)
Social cohesion index	-0.0376*** (0.0112)	-0.0219*** (0.0075)			
Male hrs. of productive work	-0.0069*** (0.0025)				
Male employed	0.0366 (0.0234)	0.0325** (0.0157)			
Severe coping adopted (LCSI)	0.0695*** (0.0235)				
Male depression (PHQ-9: 0–5)	-0.0234** (0.0099)	-0.0109* (0.0063)	0.0158** (0.0078)		
Women's diet diversity (MDDW)	-0.0185*** (0.0053)				
Food insecure HH (FCS < 28)	-0.0716*** (0.0255)				
Care work gap	-0.0072*** (0.0026)				
Male hrs. of care work	-0.0107** (0.0048)	-0.0054** (0.0027)			
Male: Twa	-0.1194* (0.0710)				
Conflict × male no schooling	0.0884* (0.0466)				0.1435** (0.0640)
Female: IDP	-0.0939* (0.0553)				
Climate shock self-reported		-0.0590*** (0.0201)			
Conflict shock self-reported		0.0388** (0.0177)			
Trust index (std.)		-0.0136* (0.0076)		0.0151* (0.0085)	
Male support for female agency (std.)		-0.0359 (0.0242)			
Female agency attitudes > p75		0.0487** (0.0192)			
Perception of attitudes from partner (female)		-0.0425 (0.0264)			
Large care gap		-0.0429** (0.0172)			
Female hrs. of productive work		-0.0033* (0.0019)			
Male: returnee			-0.0886*** (0.0333)		
Male: IDP			-0.0739* (0.0408)		
Female life satisfaction (1–7)				-0.0171** (0.0070)	
Community support for female agency (std.)				-0.0673*** (0.0242)	
HH owns or rents land				0.0314* (0.0184)	
Financial support index (std.)					-0.0252** (0.0101)
Female depression (PHQ-9: 0–5)					-0.0197* (0.0108)
Community support for fem. agency > p75					0.0467* (0.0255)
Constant	0.9273*** (0.0400)	0.3896*** (0.0254)	-0.0232 (0.0193)	0.0069 (0.0246)	0.0326 (0.0271)
Observations	1,580	1,580	1,393	1,580	1,580
R-squared	0.045	0.042	0.009	0.012	0.011

Notes: Household-level OLS with robust standard errors in parentheses; coefficients selected via forward-stepwise OLS (entry threshold $p < 0.10$); blank cells indicate the variable was not retained. Cols. (1)–(2): whether and how often partners chose different packages across the six tasks. Cols. (3)–(5): within-household female–male gaps in choosing liquidity, own-gender, and investment-oriented packages. Indices are standardized. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table A10: Model selection – alternative specifications of the choice model

	(1) Conditional logit	(2) Mixed logit all random	(3) Mixed logit transfer fixed	(4) Mixed logit transfer + sq.
Random coefficients	None	All attributes	Non-transfer	Non-transfer
Transfer specification	Dummies (3 levels)	Dummies	Dummies (fixed)	Linear + sq.
Halton draws	–	50	50	50
Observations	38,460	38,460	38,460	38,460
Parameters	11	22	19	18
Log-likelihood	-12442.8	-12250.6	-12239.5	-12244.5
AIC	24907.5	24545.2	24517.0	24525.1
BIC	25001.7	24733.4	24679.6	24679.1

LR tests against most-flexible nested benchmark (M2):

M2 vs M1: $\chi^2(11) = 384.4$, $p = 0.000$

M2 vs M3: $\chi^2(3) = -22.2$, $p = 1.000$

Notes: M1 is a conditional logit benchmark with no preference heterogeneity. M2 estimates a mixed logit with random coefficients on all attributes (including transfer-value dummies). M3 estimates a mixed logit with random coefficients on non-transfer attributes only and treats transfer-value dummies as fixed. M4 replaces the transfer dummies with a continuous transfer amount and a quadratic term to test for non-linearity. Lower AIC and BIC indicate better fit. Likelihood-ratio (LR) statistics are computed against M2; M4 is non-nested in M2 due to the change in transfer specification, so no LR test is reported.

Table A11: Attribute non-attendance – likelihood-ratio tests of constrained mixed-logit models

Constrained attribute	Log-lik.	Params.	LR stat.	df	p-value
Unrestricted (M2)	-12250.6	22	—	—	—
Transfer value (3 levels)	-12265.0	16	28.8	6	0.0001
Timing (2 levels)	-12256.2	18	11.3	4	0.0233
Scope (2 levels)	-12292.9	18	84.7	4	0.0000
Modality (2 levels)	-12353.8	18	206.5	4	0.0000
Recipient (1 level)	-12262.2	20	23.2	2	0.0000
Targeting (1 level)	-13055.6	20	1610.1	2	0.0000

Notes: Each row reports a mixed-logit model in which the indicated attribute group is constrained out of the utility specification (mean and standard-deviation coefficients dropped). The unrestricted model corresponds to specification M2 in Table A8. The likelihood-ratio statistic is $-2(\ell_R - \ell_U)$ with degrees of freedom equal to the number of constrained parameters. Rejection of the null indicates that respondents attend to the attribute when making choices.